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Synthetic Methods

SPECIAL ISSUE

Copper-Catalyzed Synthesis of Tetrasubstituted Allenes from Quaternary Ammonium Salts and Grignard Reagents

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Abstract: Although the synthesis of allenes has made considerable progress, methods for tetrasubstituted allenes are still limited. In this paper, the $CuBr_2$-catalyzed cross-coupling reaction of Grignard reagents and propargylic ammonium salts in tetrahydrofuran at 60 °C under argon delivered tetrasubstituted allenes in good to excellent yields. The reaction is thought to proceed by an S_N2' pathway.

Allenes are important units of natural products and biologically active compounds.[1–2] Moreover, they are also useful building blocks in organic synthesis as demonstrated over the last few decades.[3–17] Therefore, exploring simple and useful methods to synthesize various types of allenes is of great importance.[18] Especially, methods for the synthesis of tetrasubstituted allenes are still limited (Scheme 1).[19] In 2010, Zhang et al. developed a palladium-catalyzed three-component tandem Michael addition/cross-coupling reaction of enyne ketones with alcohols as nucleophiles and aryl halides to synthesize tetrasubstituted allenyl ketones.[19i] Davies et al. reported the reaction of propargylic alcohols with methyl diazoacetates to produce tetrasubstituted allenes through [2,3]-sigmatropic rearrangement.[19j] Sawamura's group reported a copper-catalyzed coupling between propargylic phosphates and alkyl-9-BBN to synthesize tri- and tetrasubstituted allenes (BBN = borabicyclo[3.3.1]nonane).[19k] In 2015, our group disclosed a C–H bond activation-based method to form tetrasubstituted optically active allenes, in which the C–C triple bonds of 2-alkynylic carbonates were regiospecifically inserted by an aryl–[Rh] through concerted metalation/deprotonation (CMD) followed by β-elimination.[19p]

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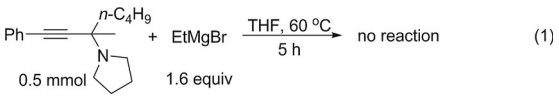
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More recently, List et al. reported a chiral disulfonimide catalyzed alkynylogous Mukaiyama aldol reaction to synthesize chiral tetrasubstituted allenoates.[19x]

Considering our previous report on KA^2 reactions (three-component coupling reaction of ketone–amine–alkyne to produce secondary propargylamines) for the efficient synthesis of propargylic amines,[20] we attempted to apply the reaction of propargylic amines and Grignard reagents to generate tetrasubstituted allenes. However, the reaction failed to yield the desired product [Eq. (1)].



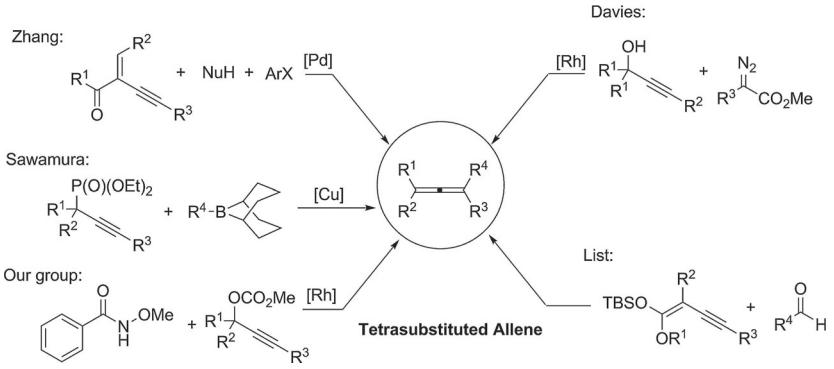
(1)

Thus, we envisioned that if we could transform such type of propargylic amines into ammonium salts, the pyrrolidinium part would be a better leaving group to form allenes (Scheme 2). To the best of our knowledge, using propargylic ammonium salts as the precursors of allenes has not been well established, and most of those transformations were achieved by reacting with $LiAlH_4$.[21] Only few reactions of propargylic ammonium salts with Grignard reagents or organocopper reagents have been reported.[22] And there are some limitations, such as narrow substrate scopes and forming alkyne as side products.

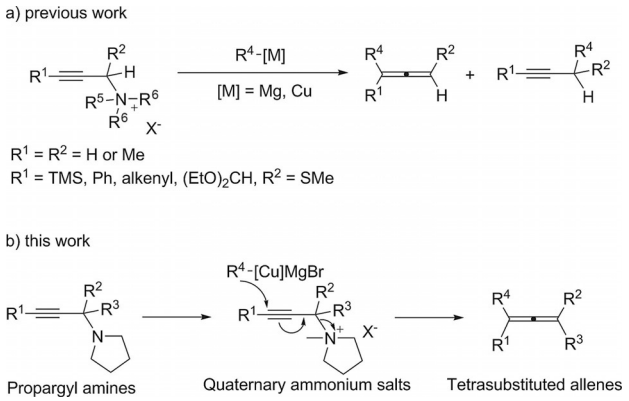
We started our investigation with quaternary ammonium salt **1a**, which was prepared through methylation with iodomethane in acetonitrile at room temperature. In the absence of any copper catalyst, the desired product **2a** was formed in 73% yield when 1.8 equivalents of ethyl magnesium bromide were added (Table 1, entry 1). After screening, it was observed that the optimal loading of ethyl magnesium bromide was 1.6 equivalents (entries 1–3). When 10 mol% of CuI was added, **2a** was afforded in 86% yield (entry 4).

The reaction of ammonium salt **1b** with ethyl magnesium bromide under the same conditions would generate the undesired trisubstituted allene product **3b** in 6% yield as the by-product (Table 2, entry 1). The employment of 10 mol% $CuBr_2$ instead of CuI led to a better selectivity (entry 2), and to our delight, **2b** was obtained in a higher yield of 90%. Increasing the amount of $CuBr_2$ could effectively reduce the formation of undesired trisubstituted allenes **3b** (entries 3–5). When using 20 mol% of $CuBr_2$, undesired product **3b** was not formed at

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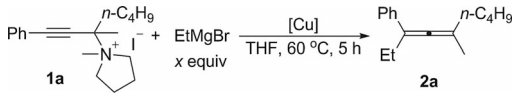


Scheme 1. Examples of the typical approaches to tetrasubstituted allenes.



Scheme 2. Substituted allenes from propargylic ammonium salts.

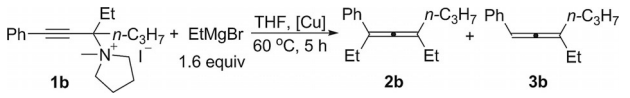
Table 1. Optimization of reaction conditions: ammonium salt **1a** with EtMgBr.[a]



Entry	[Cu] (mol %)	*x* [equiv]	Yield of **2a** [%][b]
1	–	1.8	73
2	–	1.6	72
3	–	1.4	47
4	CuI (10)	1.6	86

[a] The reaction was conducted using ammonium salt **1a** (0.5 mmol), ethyl magnesium bromide, and the copper catalyst at 60 °C in 2.5 mL of tetrahydrofuran. [b] Determined by ^{1}H NMR analysis with nitromethane as the internal standard.

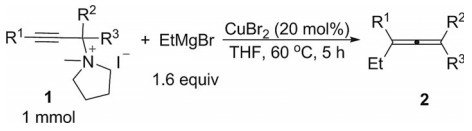
Table 2. Optimization of reaction conditions: ammonium salt **1b** with EtMgBr.[a]



Entry	[Cu] (mol %)	Yield of **2b** [%][b]	Yield of **3b** [%][b]
1[c]	CuI (10)	79	6
2	$CuBr_2$ (10)	90	4
3	$CuBr_2$ (5)	93	4
4[c]	$CuBr_2$ (15)	90 (86)[d]	1
5[c]	$CuBr_2$ (20)	90	–

[a] The reaction was conducted using ammonium salt **1b** (0.5 mmol), ethylmagnesium bromide (0.8 mmol), and the copper catalyst at 60 °C in 2.5 mL of tetrahydrofuran. [b] Determined by ^{1}H NMR analysis with nitromethane as the internal standard. [c] The reaction was conducted on 1 mmol scale. [d] Isolated yield.

Table 3. The scope of ammonium salts **1**.[a]



Entry	R^1	R^2	R^3	Yield of **2** [%][b]
1[c]	Ph	n-C_4H_9	Me (**1a**)	87 (**2a**)
2[d]	Ph	n-C_3H_7	Et (**1b**)	86 (**2b**)
3[d,e]	o-ClC_4H_6	n-C_4H_9	Me (**1c**)	86 (**2c**)
4	m-BrC_4H_6	n-C_4H_9	Me (**1d**)	80 (**2d**)
5	p-ClC_4H_6	n-C_4H_9	Me (**1e**)	79 (**2e**)
6	p-$MeOC_4H_6$	n-C_4H_9	Me (**1f**)	94 (**2f**)
7	Ph	-$(CH_2)_5$- (**1g**)		81 (**2g**)
8[d]	n-C_6H_{13}	n-C_3H_7	Et (**1h**)	88 (**2h**)

[a] The reaction was conducted using ammonium salts **1** (1 mmol), ethyl magnesium bromide (1.6 mmol), and $CuBr_2$ (20 mol%) at 60 °C in 5 mL of tetrahydrofuran. [b] Isolated yield. [c] CuI (10 mol%) and 1.8 equivalents of ethyl magnesium bromide was used. [d] $CuBr_2$ (15 mol%) was used. [e] The reaction was conducted on 0.5 mmol scale.

all. Thus, the optimal reaction conditions were defined as ammonium salts, Grignard reagents (1.6 equiv), and $CuBr_2$ (20 mol%) in tetrahydrofuran at 60 °C under argon atmosphere.

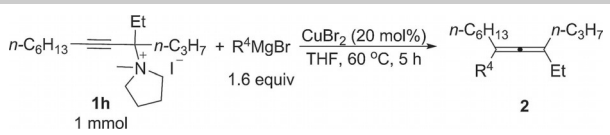
With the optimized conditions in hand, we investigated the substrate scope of the reaction. At first, different propargylic ammonium salts were tested with ethyl magnesium bromide: R^3 could be alkyl groups (Table 3, entries 1 and 2). With R^1 being an aryl group, the substrates with electron-withdrawing groups (entries 3–5) gave slightly lower yields than those with electron-donating methoxy group (entry 6). With R^2, R^3 being -$(CH_2)_5$-, tetrasubstituted allene **2g** was produced in 81% yield (entry 7). Allene **2h** with four alkyl substituents could also be synthesized through this method in 88% yield (entry 8).

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Various Grignard reagents were applicable for the method. In addition to ethyl magnesium bromide, methyl magnesium bromide and pentyl magnesium bromide also worked to afford the related products in 84% and 90% yield, respectively (Table 4, entries 1 and 2). Using freshly prepared phenyl mag-

Table 4. The scope of Grignard reagents.[a]

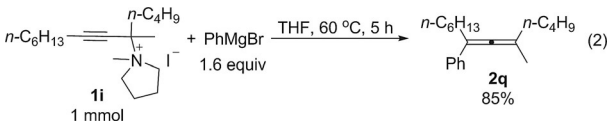


Entry	R^4	Yield of **2** [%][b]
1	Me	84 (**2i**)
2	n-C_5H_{11}	90 (**2j**)
3[c]	Ph	90 (**2k**)
4	p-ClC_4H_6	91 (**2l**)
5	p-$MeOC_4H_6$	85 (**2m**)
6[d]	m-$MeOC_4H_6$	80 (**2n**)
7[e]	Bn	89 (**2o**)
8	$CH_2{=}CHCH_2$	91 (**2p**)

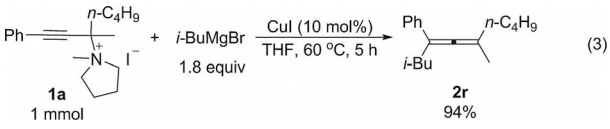
[a] The reaction was conducted using ammonium salt **1h** (1 mmol), Grignard reagents (1.6 mmol), and $CuBr_2$ (20 mol%) at 60 °C in 5 mL of tetrahydrofuran. [b] Isolated yield. [c] $CuBr_2$ (15 mol%) was used. [d] The reaction was conducted on 0.5 mmol scale. [e] Benzyl magnesium chloride was used instead of benzyl magnesium bromide.

nesium bromide, ammonium salt **1h** was converted to tetrasubstituted allene **2k** in 90% yield (entry 3). Gratifyingly, *p*-chlorophenyl magnesium bromide also worked (entry 4). On the other hand, aryl Grignard reagents with the electron-donating methoxy group underwent coupling reactions with ammonium salt **1h** successfully to afford corresponding allenes in 80–85 yields (entries 5 and 6). Benzyl magnesium chloride and allyl magnesium bromide were also suitable for this reaction (entries 7 and 8).

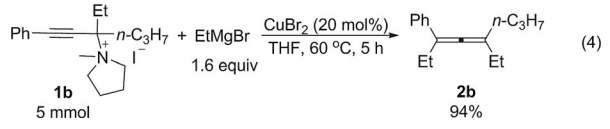
In the absence of copper catalysis, ammonium salt **1i** could also react with phenyl magnesium bromide to afford **2q** in 85% yield [Eq. (2)]. Isobutyl magnesium bromide was also applicable with the catalysis of 10 mol% CuI [Eq. (3)]. Additionally, it is convenient to conduct the reaction on a gram scale to afford **2b** in 94% yield [Eq. (4)].



(2)

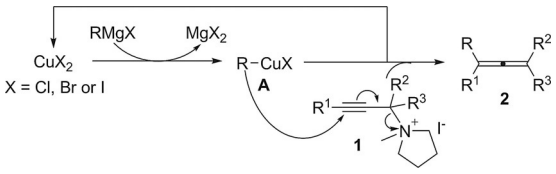


(3)



(4)

According to the previous reports,[19k,u] it is believed that a catalytic cycle starts with the transmetalation of Grignard reagents to form organocopper(II) species **A**. **A** proceeds an S_N2'-type reaction with ammonium salts **1** to generate allenes **2** (Scheme 3).



Scheme 3. A possible pathway of the reaction.

In summary, the reaction of readily available quaternary ammonium salts and Grignard reagents to produce tetrasubstituted allenes under $CuBr_2$ catalysis proceeded smoothly under mild conditions in good to excellent yields. An S_N2' mechanism was proposed. Further study on asymmetric version is ongoing in our group.

Experimental Section

Procedure for gram-scale synthesis of 2b: To a dried Schlenk tube were added $CuBr_2$ (0.3108 g, 1.4 mmol), **1b** (2.7825 g, 7 mmol), and tetrahydrofuran (35 mL) sequentially under argon atmosphere. EtMgBr (11.2 mL, 1.0 M in tetrahydrofuran, 11.2 mmol) was added dropwise with stirring. The Schlenk tube was then placed in a pre-heated oil bath at 60 °C with stirring for 5 h and monitored by TLC. After cooling to 0 °C, the crude reaction mixture was quenched with 35 mL of an aqueous saturated solution of NH_4Cl at 0 °C. The resulting mixture was extracted with diethyl ether (140 mL×3) and then dried over anhydrous Na_2SO_4. After filtration and evaporation, **2b** (1.4177 g, 94%) was afforded by chromatography on silica gel (eluent: petroleum ether) as a liquid: ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.43–7.38 (m, 2H, Ar–H), 7.29 (t, J = 7.8 Hz, 2H, Ar–H), 7.18–7.12 (m, 1H, Ar–H), 2.42 (q, J = 7.3 Hz, 2H, CH_2), 2.16–1.98 (m, 4H, 2×CH_2), 1.53–1.42 (m, 2H, CH_2), 1.11 (t, J = 7.2 Hz, 3H, CH_3), 1.03 (t, J = 7.4 Hz, 3H, CH_3), 0.92 ppm (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 200.3, 138.6, 128.1, 125.9, 125.6, 110.4, 108.4, 35.3, 26.1, 23.2, 21.2, 14.1, 12.8, 12.4 ppm.

Acknowledgements

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Keywords: allenes · ammonium salts · copper · cross-coupling reactions · Grignard reagents

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